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(54) **REAL TIME IN-SITU METROLOGY AND PROCESS PERFORMANCE IMPROVEMENT**

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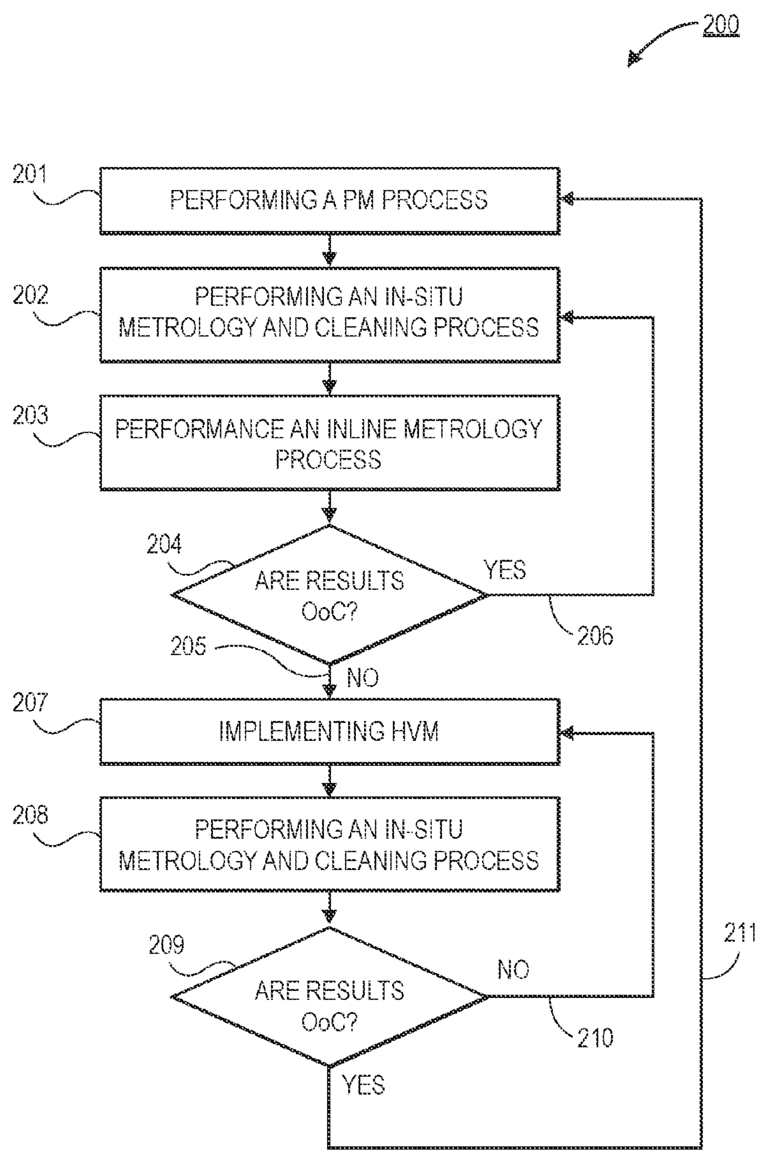
(57) **ABSTRACT**

Embodiments disclosed herein include a method for maintaining a processing tool. In an embodiment, the method comprises performing a planned maintenance (PM) process on the processing tool. The method may continue with processing substrates on the processing tool, and performing an in-situ metrology and cleaning process on the processing tool with an optimization substrate. The process may be used to determine if a processing parameter of the processing tool is within a specified range. In an embodiment, the method may continue with restarting substrate processing on the processing tool when the processing parameter is within the specified range.



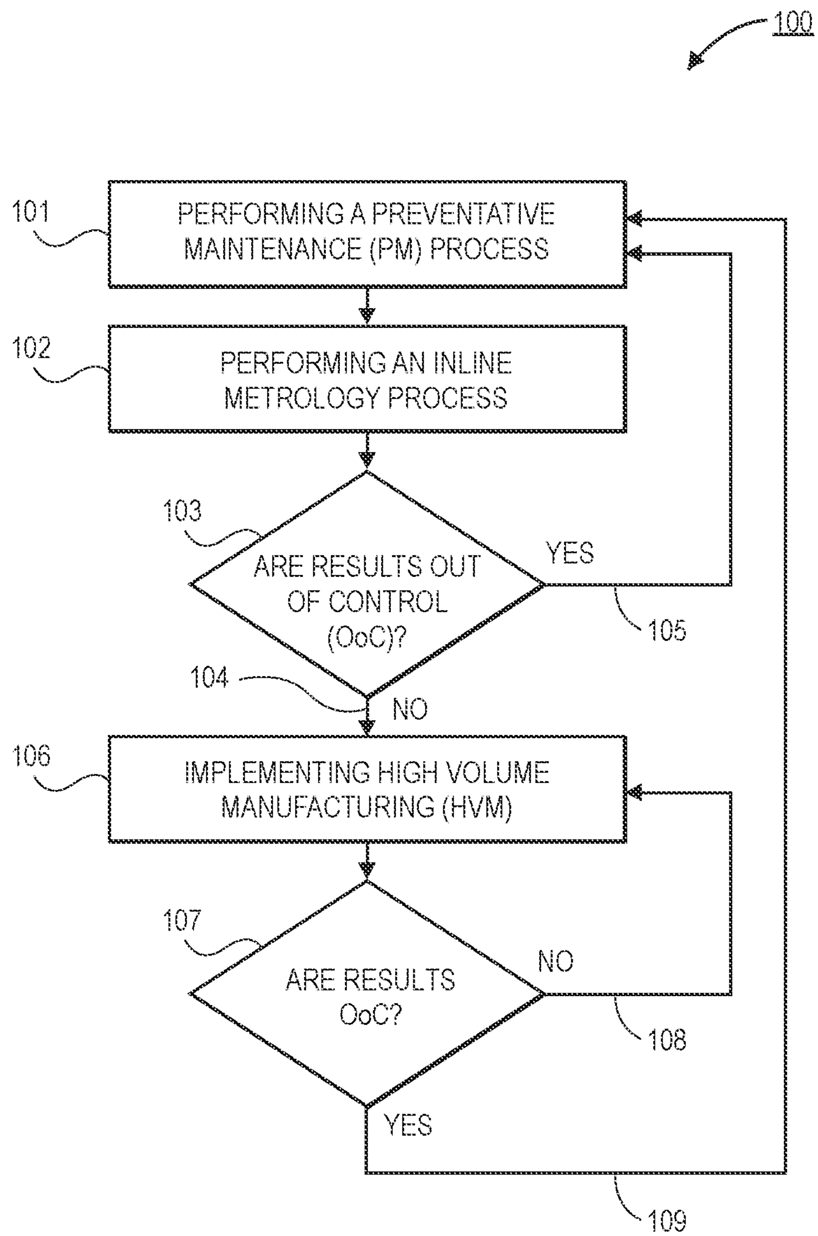

100
101
PERFORMING A PREVENTATIVE MAINTENANCE (PM) PROCESS
102
PERFORMING AN INLINE METROLOGY PROCESS
103
ARE RESULTS OUT OF CONTROL (OoC)?
YES
105
104
NO
106
IMPLEMENTING HIGH VOLUME MANUFACTURING (HVM)
107
ARE RESULTS OoC?
NO
108
YES
109

FIG. 1

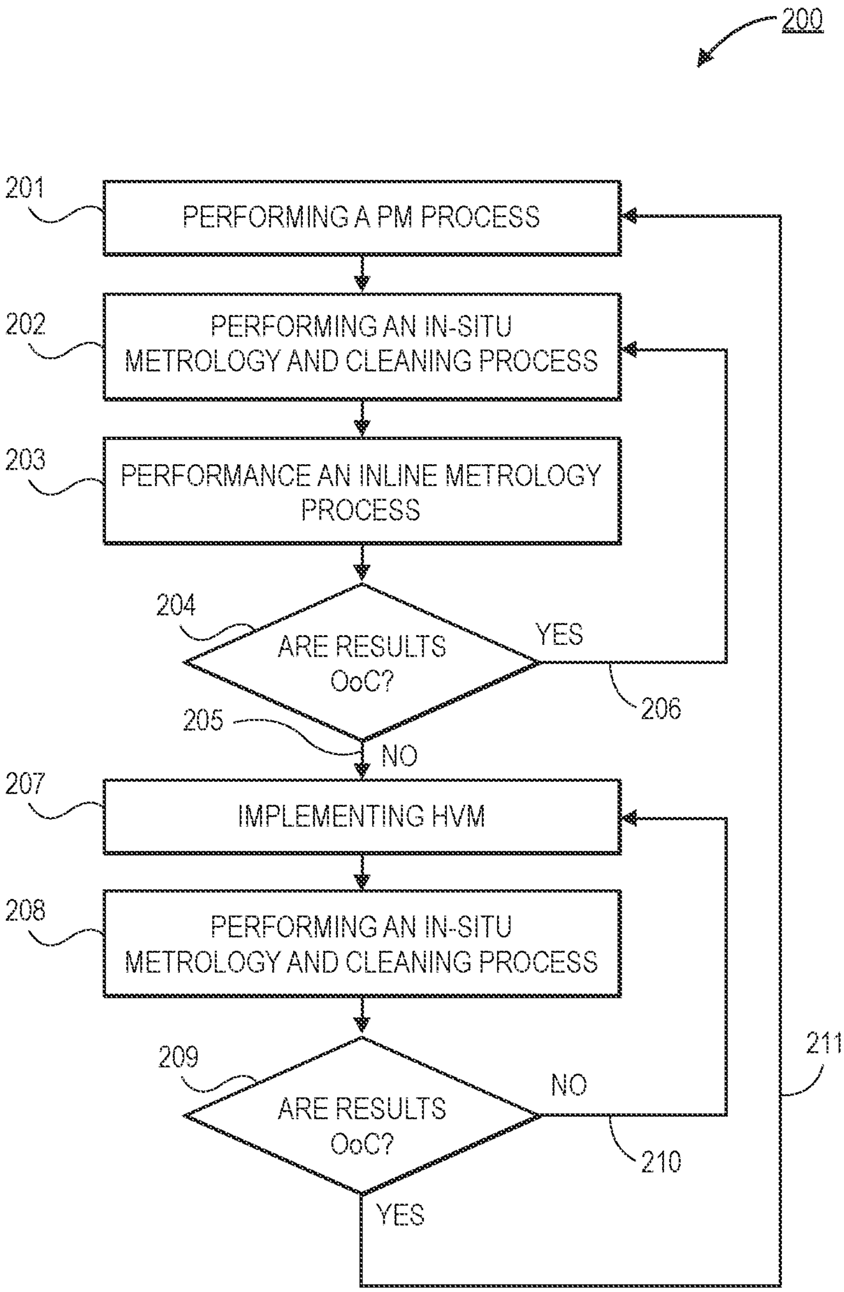

200
201
PERFORMING A PM PROCESS
202
PERFORMING AN IN-SITU METROLOGY AND CLEANING PROCESS
203
PERFORMANCE AN INLINE METROLOGY PROCESS
204
ARE RESULTS OoC?
YES
206
205
NO
207
IMPLEMENTING HVM
208
PERFORMING AN IN-SITU METROLOGY AND CLEANING PROCESS
209
ARE RESULTS OoC?
NO
210
YES
211

FIG. 2

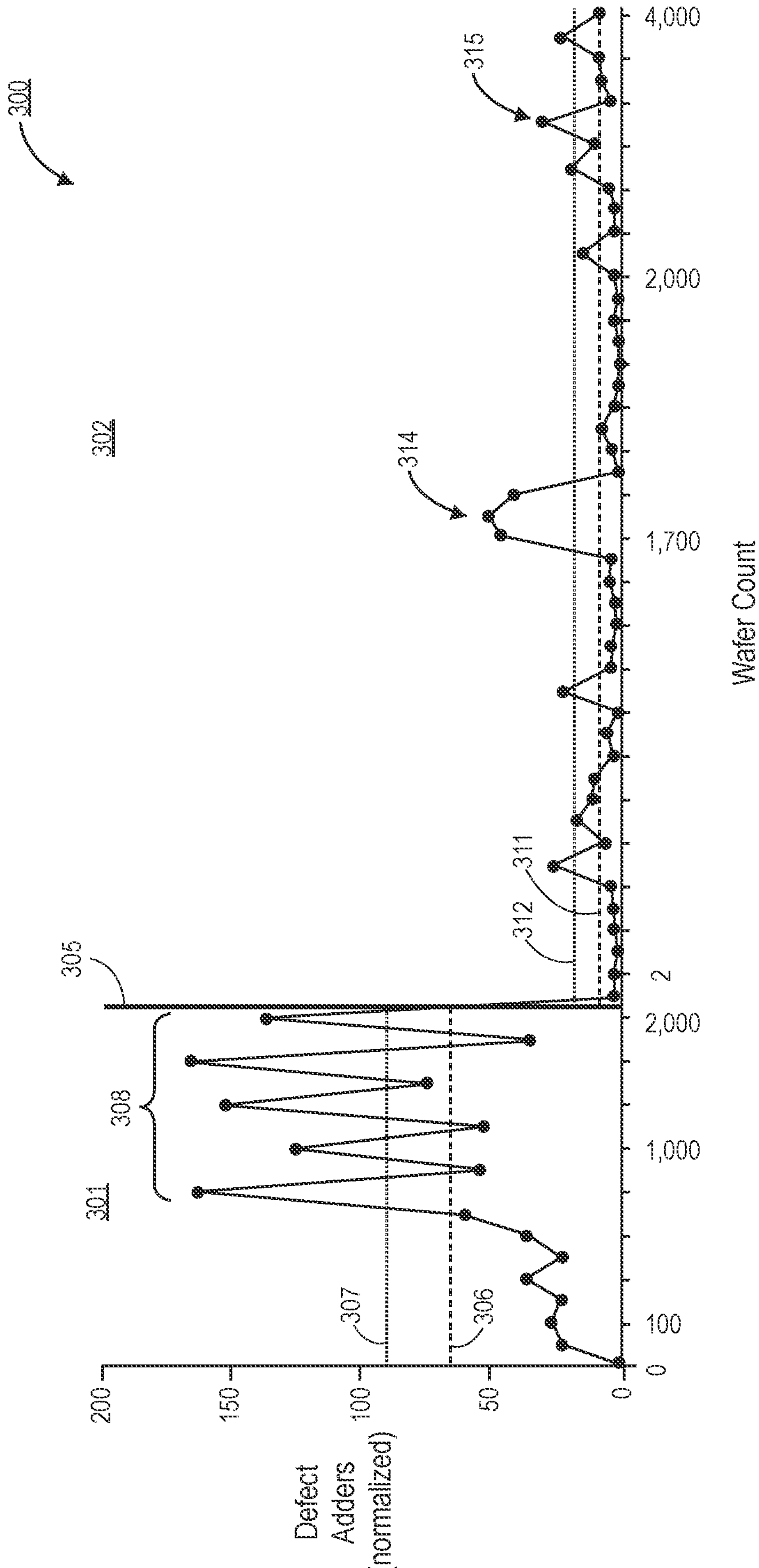

300
302
301
305
306
307
308
311
312
314
315
Defect Adders (normalized)
200
150
100
50
0
0
100
1,000
2,000
2
1,700
2,000
4,000
Wafer Count

FIG. 3

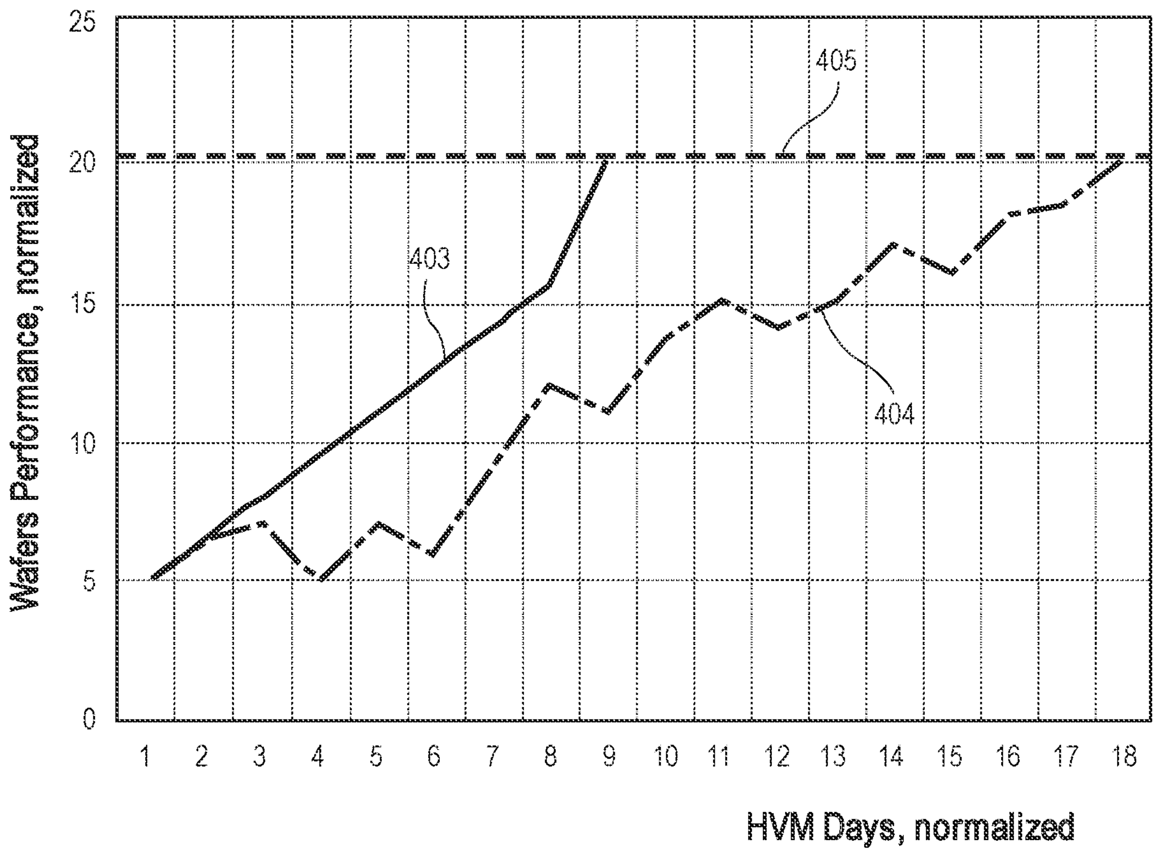

25
20
15
10
5
0
Wafers Performance, normalized
405
403
404
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
HVM Days, normalized

FIG. 4

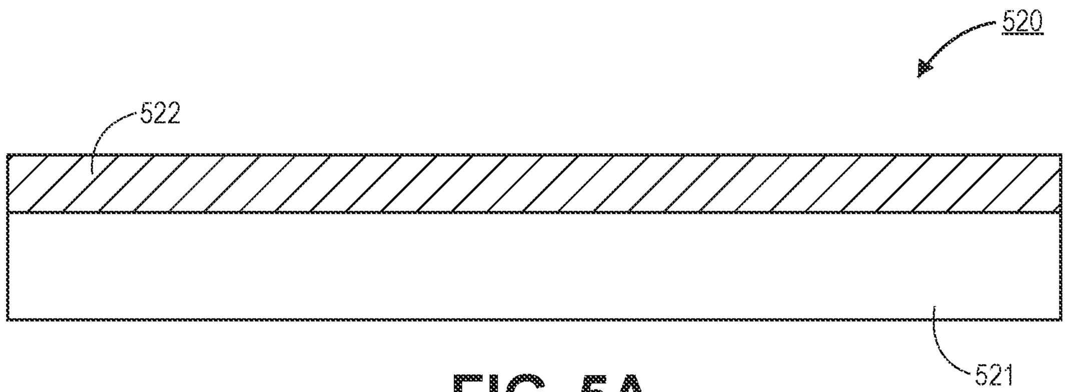


FIG. 5A

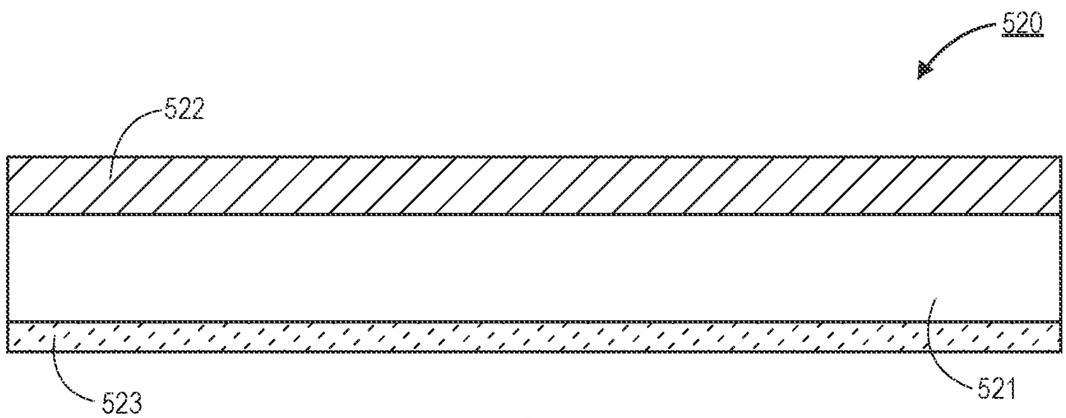


FIG. 5B

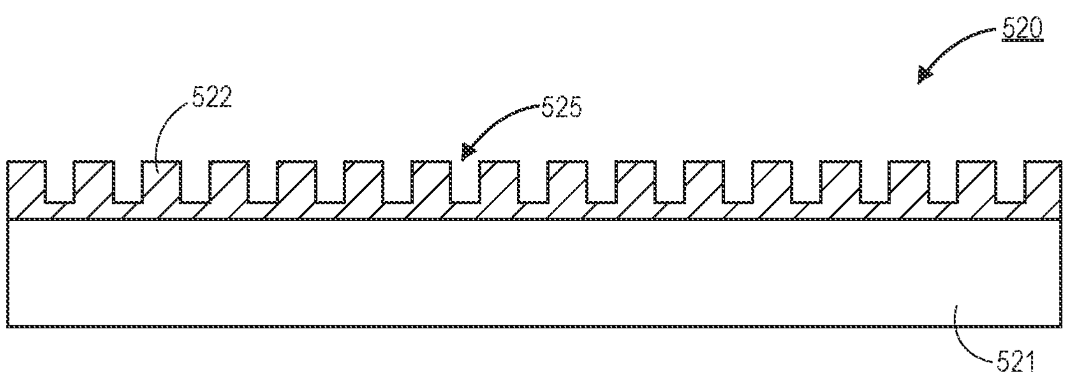


FIG. 5C

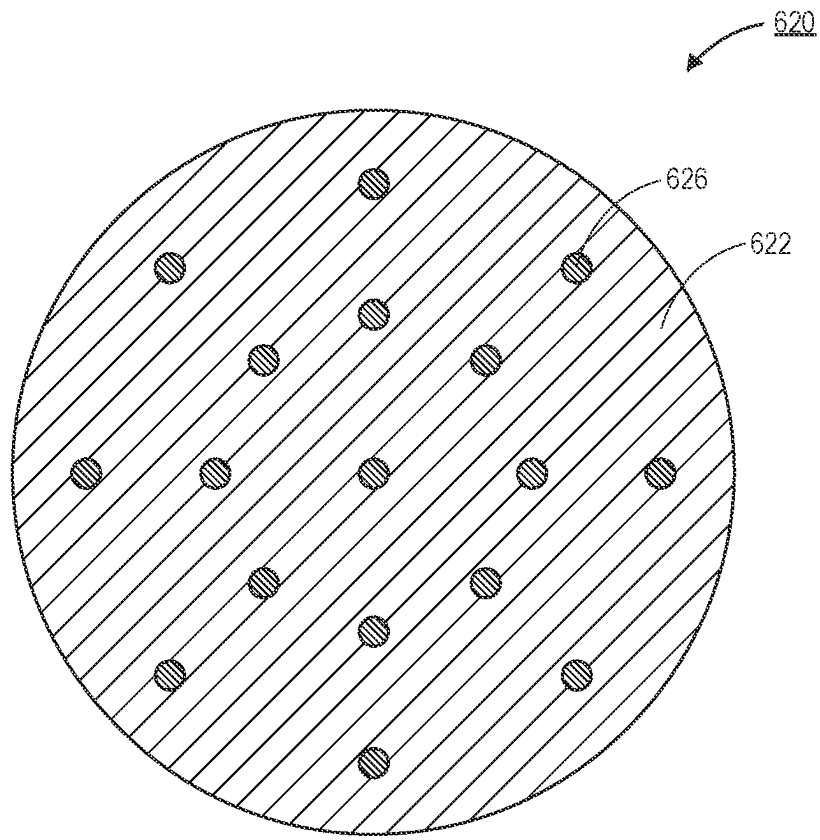


FIG. 6A

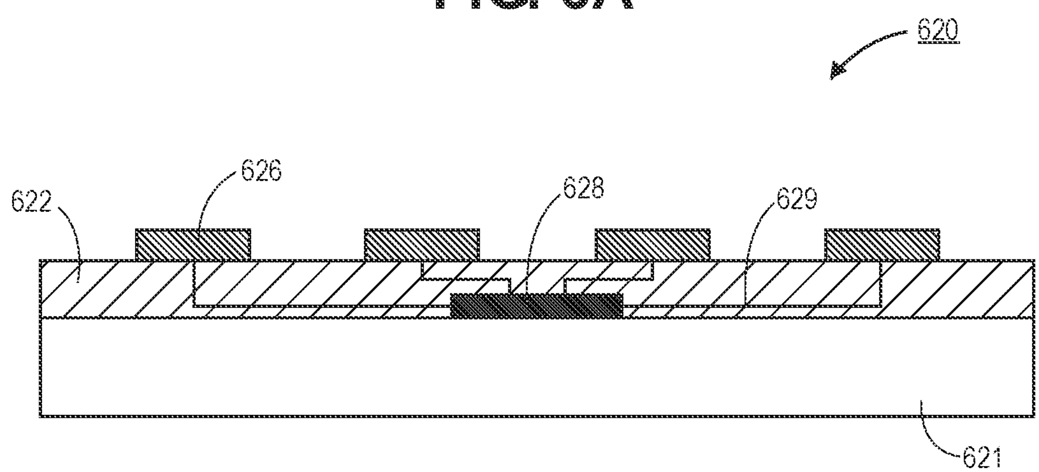


FIG. 6B

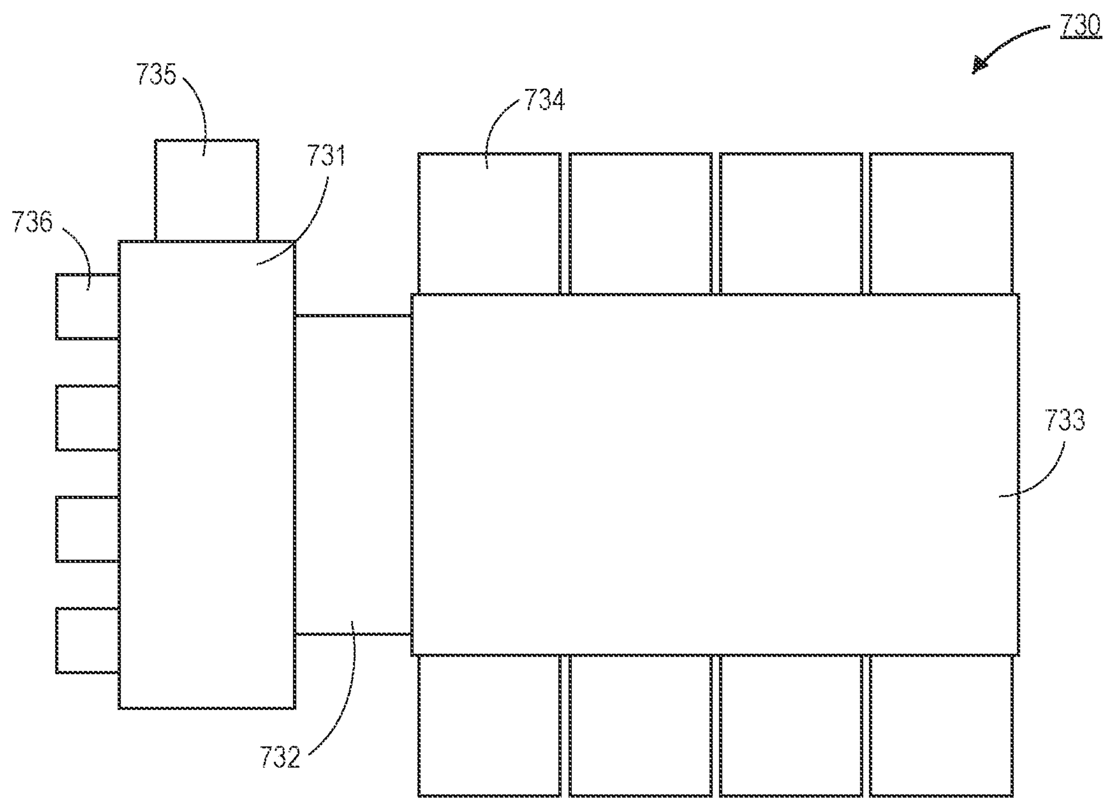


FIG. 7

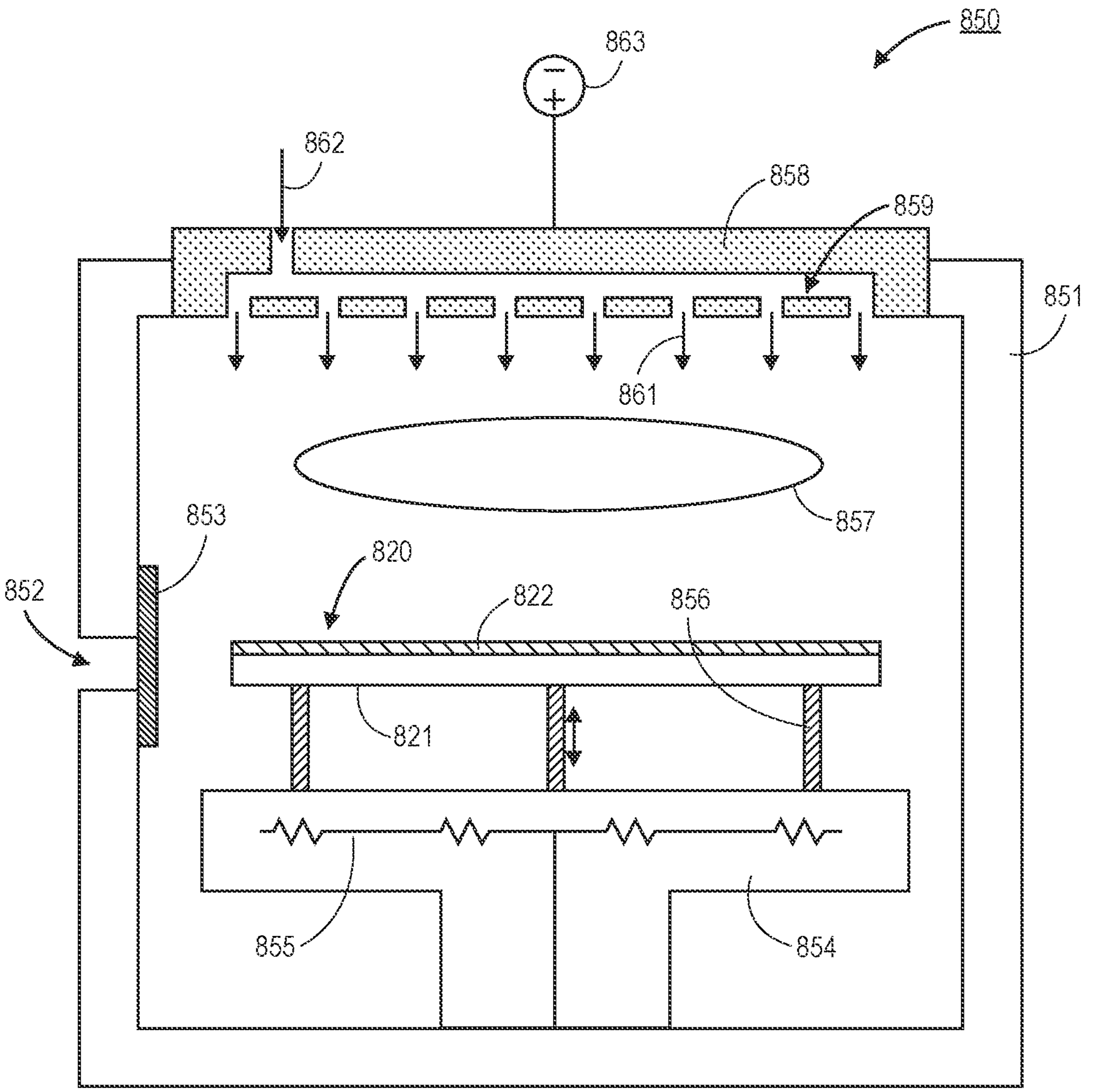


FIG. 8

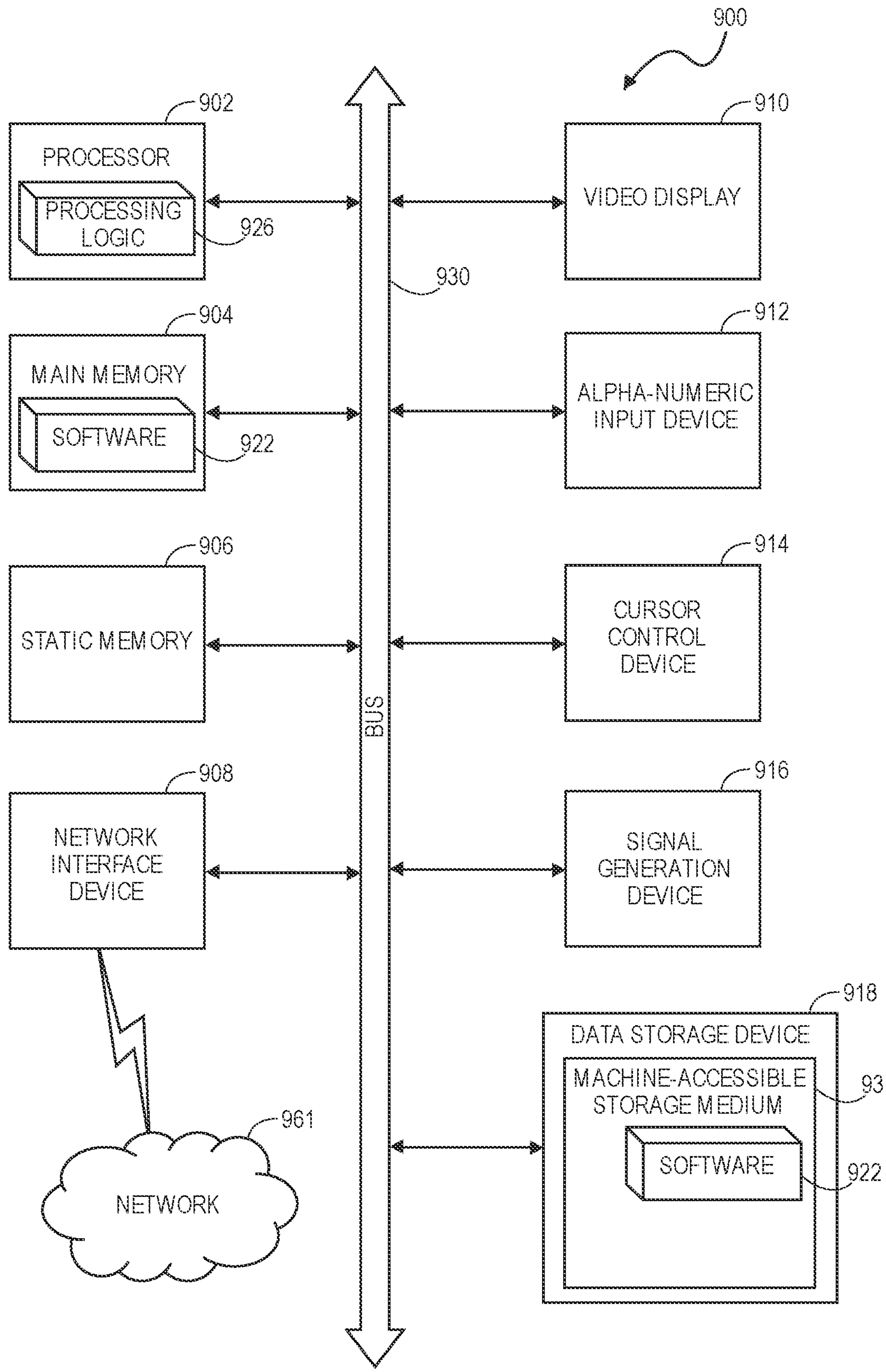

900
902
PROCESSOR
PROCESSING LOGIC
926
904
MAIN MEMORY
SOFTWARE
922
906
STATIC MEMORY
908
NETWORK INTERFACE DEVICE
961
NETWORK
930
BUS
910
VIDEO DISPLAY
912
ALPHA-NUMERIC INPUT DEVICE
914
CURSOR CONTROL DEVICE
916
SIGNAL GENERATION DEVICE
918
DATA STORAGE DEVICE
MACHINE-ACCESSIBLE STORAGE MEDIUM
931
SOFTWARE
922

FIG. 9

REAL TIME IN-SITU METROLOGY AND PROCESS PERFORMANCE IMPROVEMENT

BACKGROUND

1) Field

[0001] Embodiments relate to the field of semiconductor and/or flat panel manufacturing, in particular, to real time in-situ metrology for process improvement for semiconductor and panel fabrication processes.

2) Description of Related Art

[0002] In semiconductor manufacturing processes, smaller and more densely packed feature sizes are necessary as the size and complexity of integrated circuit structures continue to scale to smaller dimensions. In order to fabricate such features, precision and repeatability of manufacturing processes are driven to their extremes. In order to meet these requirements, the manufacturing tools, associated metrology tools, and the chambers connecting systems together need to be operating at near perfect conditions. This results in strict processing parameters that need to be met in order to continue high volume manufacturing (HVM) on a given tool.

[0003] In order to satisfy these demands, frequent maintenance is often required. Chamber cleaning recipes are a first option to bring a tool back into compliance with the strict processing parameters. However, when a chamber clean recipe is not sufficient, a planned maintenance (PM) process is implemented. PM processes may also be scheduled at standard intervals (e.g., time based or after a predetermined number of substrates have been processed).

[0004] However, PM processes are exceedingly time intensive. For example, the tool needs to be cooled, the vacuum needs to be released, and the chamber is opened. An extensive cleaning of the interior of the chamber is then completed. After the clean, the chamber will undergo a seasoning process and a qualification process is implemented to ensure the chamber is back within certain specifications. Overall, the PM process may take multiple days to complete. This significantly increases the cost of ownership of the tool, and results in a significant throughput decrease in the fabrication facility (fab).

SUMMARY

[0005] Embodiments disclosed herein include a method for maintaining a processing tool. In an embodiment, the method includes performing a planned maintenance (PM) process on the processing tool. The method may continue with processing substrates on the processing tool, and performing an in-situ metrology and cleaning process on the processing tool with an optimization substrate. The process may be used to determine if a processing parameter of the processing tool is within a specified range. In an embodiment, the method may continue with restarting substrate processing on the processing tool when the processing parameter is within the specified range.

[0006] Embodiments disclosed herein may also include an apparatus for providing in-situ metrology and chamber cleaning. In an embodiment, the apparatus includes a substrate with a first surface and a second surface opposite from the first surface. In an embodiment, a layer is on the second surface of the substrate, and the layer is configured to attract particles from an environment surrounding the apparatus. In an embodiment, a plurality of sensors are on the substrate, and the plurality of sensors are configured to detect particles landing on the apparatus. In an embodiment, the apparatus may further include a controller on the substrate that is communicatively coupled to the plurality of sensors. In an embodiment, the controller includes a processor, a memory, and a power source.

[0007] Embodiments may also include a method for maintaining a processing tool that includes performing a planned maintenance (PM) process on the processing tool, and processing substrates on the processing tool. In an embodiment, the method may further include performing an in-situ metrology and cleaning process on the processing tool with an optimization substrate to determine if a processing parameter of the processing tool is within a specified range. In an embodiment, the in-situ metrology and cleaning process includes providing the optimization substrate to the processing tool. In an embodiment, the optimization substrate includes a substrate, a layer over the substrate, that is configured to attract particles in the processing tool, and a sensor on the substrate that is configured to detect the processing parameter. In an embodiment, the method may further include processing the optimization substrate with an optimization recipe a plurality of cycles until the processing parameter is within the specified range. In an embodiment, the method may further include restarting substrate processing on the processing tool when the processing parameter is within the specified range.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a process flow diagram of a process for initiating a planned maintenance (PM) process on a tool and bringing the tool back online after the planned maintenance.

[0009] FIG. 2 is a process flow diagram of a process for initiating a PM process on a tool and bringing the tool back online with the use of an optimization substrate that executes both in-situ metrology and a cleaning process, in accordance with an embodiment.

[0010] FIG. 3 is a graph comparing the high volume manufacturing (HVM) life-cycle of a tool that uses a traditional maintenance process with a tool that uses an optimization substrate that executes both in-situ metrology and a cleaning process, in accordance with an embodiment.

[0011] FIG. 4 is a graph of the performance over time of a first tool with a traditional maintenance process and a second tool with a maintenance process that is improved through the use of an optimization substrate that executes both in-situ metrology and a cleaning process, in accordance with an embodiment.

[0012] FIG. 5A is a cross-sectional illustration of an optimization substrate with a layer for collecting particles, in accordance with an embodiment.

[0013] FIG. 5B is a cross-sectional illustration of an optimization substrate with a first layer over the substrate and a second layer under the substrate, where both the first layer and the second layer are configured to collect particles, in accordance with an embodiment.

[0014] FIG. 5C is a cross-sectional illustration of an optimization substrate with a patterned layer for collecting particles, in accordance with an embodiment.

[0015] FIG. 6A is a plan view illustration of an optimization substrate with a layer for collecting particles and a plurality of sensors, in accordance with an embodiment.

[0016] FIG. **6**B is a cross-sectional illustration of an optimization substrate with a layer for collecting particles, a plurality of sensors, and a controller that is communicatively coupled to the plurality of sensors, in accordance with an embodiment.

[0017] FIG. **7** is a plan view illustration of a cluster tool that can have an optimized maintenance schedule through the use of optimization substrates that execute both in-situ metrology and a cleaning process, in accordance with an embodiment.

[0018] FIG. **8** is a cross-sectional illustration of a chamber in a tool that can have an optimized maintenance schedule through the use of an optimization substrate that executes both in-situ metrology and a cleaning process, in accordance with an embodiment.

[0019] FIG. **9** illustrates a block diagram of an exemplary computer system that may be used in conjunction with a processing tool, in accordance with an embodiment.

DETAILED DESCRIPTION

[0020] Embodiments described herein include apparatuses and methods for real time in-situ metrology for process improvement for semiconductor and panel fabrication processes. In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments. It will be apparent to one skilled in the art that embodiments may be practiced without these specific details. In other instances, well-known aspects are not described in detail in order to not unnecessarily obscure embodiments. Furthermore, it is to be understood that the various embodiments shown in the accompanying drawings are illustrative representations and are not necessarily drawn to scale.

[0021] Various embodiments or aspects of the disclosure are described herein. In some implementations, the different embodiments are practiced separately. However, embodiments are not limited to embodiments being practiced in isolation. For example, two or more different embodiments can be combined together in order to be practiced as a single device, process, structure, or the like. The entirety of various embodiments can be combined together in some instances. In other instances, portions of a first embodiment can be combined with portions of one or more different embodiments. For example, a portion of a first embodiment can be combined with a portion of a second embodiment, or a portion of a first embodiment can be combined with a portion of a second embodiment and a portion of a third embodiment.

[0022] The embodiments illustrated and discussed in relation to the figures included herein are provided for the purpose of explaining some of the basic principles of the disclosure. However, the scope of this disclosure covers all related, potential, and/or possible, embodiments, even those differing from the idealized and/or illustrative examples presented. This disclosure covers even those embodiments which incorporate and/or utilize modern, future, and/or as of the time of this writing unknown, components, devices, systems, etc., as replacements for the functionally equivalent, analogous, and/or similar, components, devices, systems, etc., used in the embodiments illustrated and/or discussed herein for the purpose of explanation, illustration, and example.

[0023] As noted above, planned maintenance (PM) operations are necessary on semiconductor processing tools in order to meet the exceedingly precise processing parameters needed to fabricate modern integrate circuitry and the like. However, the PM processes as they exist today are implemented at intervals that make cost of ownership of the tool high. Repeated PM processes also decrease throughput in the fabrication facility (fab). The frequency of the PM processes is driven (at least in part) by the constant drift of the processing parameters once the tool has been qualified. Typically, this drift tends to bring the tool out of control (OoC) (also referred to as being out of specification, or out of compliance) over time.

[0024] One particular parameter that can lead to defects is particle generation. When particles land on portions of circuitry, the particle can completely ruin (i.e., render defective) a device. This can lead to significant yield reductions. As such, when particle concentrations within a tool get too high, the tool is considered OoC. Particles can be generated by many different sources. For example, friction between a substrate and one or more of a pedestal, a robot, a cover ring, etc., may result in the generation of particles. Particles may also be introduced into the tool from carry on gasses and/or precursor gasses. Plasma and/or chemical reactions may also generate particles. Accordingly, as more substrates are processed the particle concentration will generally increase.

[0025] Referring now to FIG. **1**, a process **100** for performing a PM process, qualifying the tool, and operating the tool in an high volume manufacturing (HVM) condition is shown using typical processes. As shown, the process **100** may include operation **101**, which comprises performing a PM process on the tool. The PM process may include cooling the tool, releasing a vacuum within a chamber, and opening the chamber. An extensive cleaning of the interior of the chamber is then completed. After the clean, the chamber will undergo a seasoning process.

[0026] The seasoning process may include running one or more qualifying substrates to determine if the tool is meeting specifications. The qualifying substrates are non-production substrates that are used for the sole purpose of verifying the performance of the tool. The number of qualifying substrates that are run may be as few as one or as many as twenty or more. All qualifying substrates may undergo the same process recipe. Alternatively, qualifying substrates may undergo two or more different process recipes in order to quantify different process regimes.

[0027] After operation **101**, the process **100** may continue with operation **102**, which comprises performing an inline metrology process. The inline metrology may be performed on one or more of the qualifying substrates. The inline metrology process may be referred to as a non in-situ metrology process. That is, the metrology is performed outside of the processing chamber. The inline metrology process may include a defect detection process. For example, an incident laser beam directed at the substrate may be used to detect imperfections on the substrate surface. That is, a defect may be detected when the laser beam is scattered.

[0028] Following operation **102**, the process **100** may continue to decision block **103** where it is determined if the results of the inline metrology indicate that the tool is OoC. If the YES branch **105** is taken (i.e., the tool is OoC), then the process **100** loops back to operation **101**, and another PM process is performed. This is a significant event since PM processes are long and expensive events. However, under typical process flows, the chance of requiring a second PM process is not insignificant. This is due (at least in part) to the PM process being largely a manual process and is susceptible to human error. Furthermore, another set of qualifying substrates are processed and additional inline metrology (operation **102**) is needed when branch **105** is taken.

[0029] After the tool is qualified (i.e., the NO branch **104** is taken), the tool is brought online and operation **106** is started, which comprises implementing HVM production. During HVM production, production substrates may be chosen for further inline metrology at predetermined intervals (e.g., based on time or based on number of substrates processed). At each instance of inline metrology, decision block **107** is implemented to determine if the results indicate the tool is OoC. In the desirable case, the NO branch **108** is taken, and the process **100** loops back to operation **106**, where HVM production is continued. However, if the YES branch **109** is taken, the process **100** loops back to operation **101**, and an additional PM process is started.

[0030] It is to be appreciated that once operation **106** is initiated (i.e., HVM production), some in-situ chamber clean recipes may be implemented, but otherwise, the tool is left to drift towards being OoC. Depending on the particular tool, this may lead to a relatively short period of HVM production (e.g., several days to a week). When viewed in light of the extensive requalification process that may include one or more PM processes, this limited uptime can lead to high cost of ownership of the tool and decreased throughput in the fab.

[0031] Accordingly, embodiments disclosed herein include the use of an optimization substrate. The optimization substrate may include the functionality to provide in-situ metrology while also cleaning the tool at the same time. For example, in the case of the monitoring and control of particle generation, the optimization substrate may include a layer configured to attract and trap particles in order to reduce the particle concentration in the tool. The optimization substrate may also include one or more sensors for measuring the particle concentration within the tool.

[0032] Such an optimization substrate can benefit a tool in several ways. One benefit is in the qualification process. By running an optimization substrate through the tool one or more times before qualification, the inline metrology is more likely to yield passing results. This can result in the ability to move to HVM production without additional PM processes. Additionally, during the HVM production, optimization substrates can be cycled at certain intervals to monitor the targeted process parameter (e.g., particle concentration) while also improving the process parameter (e.g., removing particles from the tool). This can lengthen the uptime of the tool, which reduces cost of ownership and improves throughput in the fab.

[0033] While particle generation is one issue that can render a tool OoC, other processing parameters may also result in non-optimal tool conditions. For example, plasma non-uniformity (e.g., with respect to density, composition, etc.), UV uniformity, excessive vibrations, temperature uniformity (e.g., uneven heat flux), gas flow uniformity, chamber calibration (e.g., chamber matching between chambers), and the like can all lead to a tool being OoC.

[0034] In the embodiments described herein, particular explanation is given to the issue of particle generation, but it is to be appreciated that many different processing parameters may be addressed with embodiments disclosed herein. For example, when an optimization substrate is described with sensors, the sensors may include sensors suitable for detecting any of the processing parameter conditions (or sensors for detecting conditions related to those processing parameter conditions).

[0035] Additionally, multiple different sensor types can be provided on a single optimization substrate. For example, particle detection sensors and temperature sensors may both be provided on the optimization substrate. This allows for the detection of both particle concentration excursions and temperature excursions with a single optimization substrate.

[0036] In some embodiments, the process parameter sensor may also be paired with a position sensor. For example, particle detection sensors may be paired with one or more accelerometers on a single optimization substrate. This may allow for position tracking of the optimization substrate so that the location of the optimization substrate is known when particles are detected. This can be used to find regions or locations within the tool where high levels of particles are located. Further cleaning (e.g., by allowing the optimization substrate to reside in those locations for longer durations) may then be implemented to prolong the uptime of the tool. When a PM is ultimately needed, the identified locations of high particle concentration can be given special attention during the cleaning process.

[0037] In an embodiment, any type of tool may benefit from the use of an optimization substrate such as those described herein. For example, tools such as, but not limited to, physical vapor deposition (PVD) tools, atomic layer deposition (ALD) tools, chemical vapor deposition (CVD) tools, plasma enhanced ALD or CVD tools, etching tools, front end packaging (FEP) tools, electrochemical plating (ECP) tools, ion implantation tools, or the like may be used in accordance with embodiments described herein. Metrology tools (e.g., laser particle detection tools, scanning electron microscopy (SEM) tools, critical dimension SEM (CD-SEM) tools, bright and dark optical inspection tool, or the like) may also benefit from optimization substrates. In an embodiment, the tool may also comprise a cluster tool with one or more different types of chambers. The optimization tool may provide validation and cleaning to any of the chambers, transfer chambers, load locks, equipment front end modules (EFEMs), or the like.

[0038] In an embodiment, the optimization substrate may take any suitable form factor. For example, the optimization substrate may have a wafer-like form factor. That is, the optimization substrate may be roughly circular with a diameter of 300 mm, 450 mm, or the like. A thickness of the optimization substrate may allow for transport within the tool under investigation using existing wafer handling robots and/or the like. In other embodiments, the optimization substrate may have a panel form factor, a half-panel form factor, a quarter-panel form factor, or the like. That is, embodiments may include wafer based tools for integrated circuit processing or panel based tools for electronics packaging, display technologies, and/or the like.

[0039] Referring now to FIG. **2**, a process **200** for performing a PM process, qualifying the tool, and operating the tool in an HVM condition is shown, in accordance with an embodiment. In an embodiment, the process **200** may include operation **201**, which comprises performing a PM process on the tool. The PM process may include cooling the tool, releasing a vacuum within a chamber, and opening the chamber. An extensive cleaning of the interior of the chamber is then completed. After the clean, the chamber may undergo a seasoning process.

[0040] In an embodiment, the tool may comprise any type of tool, such as those described in greater detail herein. For example, the tool may include a deposition tool, an etching tool, an ion implantation tool, or a metrology tool. The tool may also be a cluster tool that comprises one or more different types of chambers that are coupled together with transfer chambers, load locks, EFEMs, and/or the like. The process **200** may be used in order to qualify and/or maintain a single chamber within a cluster tool, multiple different chambers within a cluster tool, and/or the entire cluster tool.

[0041] In an embodiment, the process **200** may then continue with operation **202**, which comprises performing an in-situ metrology and cleaning process. In an embodiment, the in-situ metrology and cleaning process may be implemented by providing an optimization substrate into the tool. The optimization substrate may include one or more sensors for detecting and/or determining a process parameter (e.g., particle concentration, thermal properties, plasma properties, vibration, etc.), and a surface for cleaning the tool (e.g., a surface configured to trap particles). In an embodiment, the optimization substrate may be treated before being sent into the tool. For example, the optimization substrate may be charged (with a positive charge or a negative charge) and/or heated. Providing a charge and/or heating the optimization substrate may improve the efficiency of particle capture in some embodiments.

[0042] In an embodiment, operation **202** may include running the optimization substrate through the tool while implementing an optimization recipe. The optimization recipe may include setting one or more of gas flow rates, duration within the tool, positioning within the tool, temperature cycles, plasma cycles, lift pin cycles, and/or the like to certain values in order to aid in the collection of data and/or to improve particle collection. Operation **202** may include running the optimization substrate through the tool with the optimization recipe a single time, or multiple cycles may be implemented.

[0043] As opposed to going directly to the qualification process, as described in process **100** above, the inclusion of operation **202** allows for the tool to be cleaned further and for initial metrology measurements to be taken. Particularly, the use of in-situ metrology measurement (i.e., metrology measurements within the chamber during operation) allow for quicker analysis of the cleanliness of the tool. The optimization substrate can be cycled through the tool any number of times in operation **202** in order to improve the probability of passing the qualification process. For example, operation **202** may be repeated until the in-situ metrology reports an acceptable level for particle concentration. Furthermore, each cycle will reduce the particle concentration due to the cleaning effects of the optimization substrate.

[0044] After operation **202** is completed, qualifying substrates may be run through the tool with a process similar to the use of qualifying substrates described above. The qualifying substrates may then be used in operation **203**, which comprises performing an inline metrology process. Following operation **203**, the process **200** may continue to decision block **204** where it is determined if the results of the inline metrology indicate that the tool is OoC. If the YES branch **206** is take (i.e., the tool is OoC), then the process **200** loops back to operation **202**, and one or more cycles of in-situ metrology and cleaning is implemented.

[0045] After the tool is qualified (i.e., the NO branch **205** is taken), the tool is brought online and operation **207** is started, which comprises implementing HVM production. In an embodiment, the path to get to operation **207** is improved compared to the process **100** described in greater detail above. This is because the probability of needing additional PM processes is significantly reduced. This is for several reasons. First, the in-situ metrology and cleaning process enhances the performance of the tool, and allows for moving on to the qualifying process only after certain metrics are hit (e.g., particle concentration thresholds). This makes it more likely that the tool is within specification after the first qualifying process. Additionally, even if the qualifying process fails, additional cycles of the in-situ metrology and cleaning process can be implemented to help move the tool towards qualification. However, if multiple failures occur at decision block **204**, the process may proceed back and implement another PM process. Though, this will be rare, and fewer PM processes will be needed during the life of the tool compared to the use of process **100**.

[0046] In an embodiment, process **200** may continue with operation **208**, which comprises performing an in-situ metrology and cleaning process. In an embodiment, the in-situ metrology and cleaning process may be implemented by providing an optimization substrate into the tool. The optimization substrate may include one or more sensors for detecting and/or determining a process parameter (e.g., particle concentration, thermal properties, plasma properties, vibration, etc.), and a surface for cleaning the tool (e.g., a surface configured to trap particles). In an embodiment, the optimization substrate may be treated before being sent into the tool (e.g., charged and/or heated). Multiple cycles of the in-situ metrology and cleaning process may also be used in some embodiments. More generally, operation **208** may be similar to operation **202** in some embodiments.

[0047] In an embodiment, operation **208** may be implemented after any suitable duration after the beginning of operation **207**. In some embodiments, operation **208** may occur after a certain number of substrates have been processed (e.g., 50 substrates or more, 100 substrates or more, 500 substrates or more). In other embodiments, operation **208** may occur after a suitable period of time (e.g., after 12 hours, after 24 hours, after 48 hours, etc.). The timing of operation **208** may also be chosen to coincide with a time period when the tool is otherwise idling. In such instances, useful production time is not used by the inline metrology and cleaning process.

[0048] In an embodiment, operation **208** provides a couple benefits. First, the in-situ metrology component allows for monitoring a parameter of the chamber (e.g., particle concentration). This allows for a picture of tool health to be obtained in order to help determine when the next PM process is necessary. Second, the cleaning component improves the longevity of the HVM duration. For example, removing particles from the tool allows for the tool to remain within specification for a longer duration. This provides a longer time period between PM events, which reduces cost of ownership and improves fab throughput.

[0049] In an embodiment, process **200** may continue to decision block **209** after operation **208**. At decision block **209**, it is determined if the tool is OoC. This decision can be based (at least in part) on the results of the in-situ metrology from operation **208**. If the tool is not OoC, the NO branch **210** is taken, and the process **200** returns to operation **207** so that HVM production can continue. If the tool is OoC, the YES branch **211** is taken. This may result in the process **200** returning back to operation **201**, where a new PM process is started. In some embodiments, if the YES branch **211** is taken, an inline metrology analysis may be implemented to confirm the finding before returning to operation **201**.

[0050] Referring now to FIG. **3**, a graph **300** illustrating a count of defect adders over the course of processing substrates is shown, in accordance with an embodiment. The first section **301** of the graph to the left of line **305** illustrates a tool that uses a process similar to process **100** described in greater detail above. As shown, the defect count becomes unstable in region **308** with defect counts greatly exceeding the warning bands **306** to **307** that set the OoC specification of the tool. As such, a PM process is quickly needed.

[0051] In contrast the second section **302** to the right of the line **305** illustrates a tool that uses a process similar to process **200** described in greater detail above. As shown, the warning bands **311** and **312** can be set lower, and the consistency of the operation is improved. As shown, when high defect adder counts are detected (e.g., at location **314** and location **315**), one or more cycles using an optimization substrate can be run in order to monitor (with in-situ metrology) and clean the tool. As such, the duration of HVM operation can be extended. For example, FIG. **3** shows up to 4,000 substrates being processed (which is twice as many as in the first section **301**), and the tool is still within specification in order to continue with HVM processes.

[0052] Referring now to FIG. **4**, a graph showing wafer performance over the course of HVM days (i.e., days during which HVM processing is implemented) is shown, in accordance with an embodiment. As shown, the threshold **405** for being OoC or out of specification is around 20. Once the performance exceeds this predetermined level, the tool will need a PM process in order to bring the tool back into compliance with the specification.

[0053] As shown, the line **403** showing a tool operating under process **100** is continuously increasing. This corresponds to the tool drifting towards being out of specification during the entire HVM run. In contrast, the line **404** showing a tool operating under process **200** includes increasing and decreasing portions. The decreasing portions correspond to use of an optimization substrate (either a single cycle or multiple cycles). Accordingly, by improving the wafer performance at regular intervals, the HVM duration can be increased. For example, the HVM duration of line **403** is 8.75 days, and the HVM duration of line **404** is approximately 17.75 days (which is over double the duration of line **403**). Accordingly, the use of process **200** can significantly increase tool uptime, which reduces cost of ownership and increases throughput in the fab.

[0054] Referring now to FIGS. **5A-5C**, a series of cross-sectional illustrations depicting portions of an optimization substrate **520** is shown, in accordance with various embodiments.

[0055] Referring now to FIG. **5A**, a cross-sectional illustration of the optimization substrate **520** is shown, in accordance with an embodiment. In an embodiment, the optimization substrate **520** may comprise a substrate **521**. The substrate **521** may have a first surface (i.e., bottom surface) and a second surface (i.e., top surface). In an embodiment, a layer **522** is provided over the second surface. In an embodiment, the layer **522** may be a blanketed layer. That is, the layer **522** may cover substantially all of the second surface of the substrate **521**. Though, in other embodiments, the layer **522** may cover a portion of the second surface of the substrate **521**.

[0056] In an embodiment, the layer **522** may be a material (or materials) that is configured to attract and/or retain particles from a tool environment. The attraction of particles to the layer **522** may be improved by providing a charge (e.g., positive or negative) to the optimization substrate **520** and/or by heating the optimization substrate **520** before being provided to the tool. In other embodiments, a surface roughness of the layer **522** may be relatively high in order to provide a greater surface area for attracting and retaining particles.

[0057] In an embodiment, the optimization substrate **520** may have a wafer-like form factor. That is, the optimization substrate may be roughly circular with a diameter of 300 mm, 450 mm, or the like. A thickness of the optimization substrate (e.g., including the substrate **521** and the layer **522**) may allow for transport within the tool under investigation using existing wafer handling robots and/or the like. For example, a total thickness of the optimization substrate **520** may be approximately 5 mm or less, approximately 3 mm or less, or approximately 1 mm or less. In other embodiments, the optimization substrate **520** may have a panel form factor, a half-panel form factor, a quarter-panel form factor, or the like. That is, embodiments may include wafer based tools for integrated circuit processing or panel based tools for electronics packaging, display technologies, and/or the like.

[0058] In an embodiment, the substrate **521** may comprise any material suitable for transport within a tool. For example, the substrate **521** may have a modulus and/or rigidity that enables transferring the optimization substrate **520** with substrate handling robots within the tool. For example, the substrate **521** may comprise silicon, silicon carbide, graphite, aluminum, a ceramic, or any other suitable dielectric or conductive materials. In an embodiment, the layer **522** may comprise one or more of silicon nitride, silicon oxide, alumina, aluminum, titanium, a polymer, or a sol-gel (e.g., a silicon oxide sol-gel, an alumina sol-gel, or the like).

[0059] Referring now to FIG. **5B**, a cross-sectional illustration of an optimization substrate **520** is shown, in accordance with an additional embodiment. The optimization substrate **520** in FIG. **5B** may be similar to the optimization substrate **520** in FIG. **5A**, with the addition of a second layer **523** on the first surface (i.e., bottom surface) of the substrate **521**. In an embodiment, the second layer **523** may comprise the same material (or materials) as the layer **522**. In other embodiments, the second layer **523** may comprise a different material than the layer **522**. Material options for the second layer **523** may be similar to those listed above for use in the layer **522**. Additionally, a thickness of the second layer **523** may be substantially equal to a thickness of the layer **522**, or a thickness of the second layer **523** may be different than a thickness of the layer **522**. The second layer **523** may be a blanket layer across the entire first surface, or the second layer **523** may cover a portion of the first surface.

[0060] In an embodiment, providing a second layer **523** on the optimization substrate **520** allows for a greater area to accumulate particles, which improves the cleaning efficiency of the optimization substrate **520**. This additional area may be exposed as the optimization substrate **520** is passed through the tool (e.g., by substrate handling robots and/or the like).

[0061] Referring now to FIG. **5**C, a cross-sectional illustration of an optimization substrate **520** is shown, in accordance with yet another embodiment. In an embodiment, the optimization substrate **520** in FIG. **5**C may be similar to the optimization substrate **520** in FIG. **5**A, with the addition of a pattern **525** in the layer **522**. The pattern **525** may comprise one or more trenches or other topography at the surface of the layer **522**. In the illustrated embodiment, the pattern **525** has a plurality of uniform trenches. Though, trench dimensions (e.g., depth, width, length, slope of sidewalls, etc.) may be non-uniform across the layer **522**. In an embodiment, the pattern **525** may increase the surface area of the layer **522** in order to improve attraction and capture of particles. While no second layer is shown in FIG. **5**C on the first surface (i.e., bottom surface) of the substrate **521**, embodiments may also comprise a second layer (patterned or unpatterned) on the first surface of the substrate **521**.

[0062] Referring now to FIGS. **6**A and **6**B, illustrations of portions of an optimization substrate **620** that illustrate the inclusion of one or more sensors **626** is shown, in accordance with an embodiment.

[0063] Referring now to FIG. **6**A, a plan view illustration of the optimization substrate **620** is shown, in accordance with an embodiment. As shown, a layer **622** for capturing particles is provided as a top surface of the optimization substrate **620**. The layer **622** may be similar to any of the layers **522** described in greater detail above. In an embodiment, one or more sensors **626** may be distributed across a surface of the optimization substrate **620**. The sensors **626** may be sensors configured for providing the in-situ metrology operations with respect to process **200** described in greater detail herein. In an embodiment, the sensors **626** may be provided with any suitable layout and/or pattern across the optimization substrate **620**. For example, sensors **626** may be arranged in a set of two concentric circles with an additional sensor **626** at a center of the optimization substrate **620**.

[0064] In an embodiment, the sensors **626** may be sensors suitable for performing any suitable type of in-situ metrology. In a first embodiment, the sensors **626** may comprise particle sensors. For example, the sensors **626** may comprise a vibration sensor, a resonant structure sensor (e.g., a cantilever structure, or the like), or a laser light scattering sensor. Embodiments, may also comprise sensors for detecting vibrations, temperature, gas flow rates, optical stimulus (e.g., a camera, an optical emission spectroscopy (OES) system for determining plasma properties), and/or the like.

[0065] In some embodiments, the sensors **626** may also comprise one or more sensors for determining a position of the optimization substrate **620** within the tool. For example, an accelerometer (e.g., a three-axis accelerometer) may be used for position tracking. Position tracking may be beneficial since correlating a position of the optimization substrate **620** with particle detection can identify locations where particle concentrations are higher. This can inform the process where to place an optimization substrate **620** within a tool for cleaning and/or inform a decision on how long an optimization substrate **620** should remain at a certain location.

[0066] In an embodiment, a single type of sensor **626** may be provided on the optimization substrate **620**. In other embodiments, two or more different types of sensors **626** may be provided on a single optimization substrate **620**. In such an embodiment, the in-situ metrology process can track and/or monitor multiple different process parameters at the same time. When a single type of sensor **626** is used, multiple different optimization substrates **620** (each with different types of sensors **626**) may be cycled through the tool to track and/or monitor multiple different process parameters.

[0067] Referring now to FIG. **6**B, a cross-sectional illustration of a portion of an optimization substrate **620** is shown, in accordance with an embodiment. In an embodiment, the sensors **626** are provided over the surface of the layer **622**. In other embodiments, the sensors **626** may be provided on the substrate **620**, partially embedded in the layer **622**, or provided in any suitable location. Further, while the sensors **626** are only shown as being over the top surface of the substrate **621**, embodiments may also include one or more sensors **626** over the bottom surface of the substrate **621**.

[0068] In an embodiment, the sensors **626** may be communicatively coupled to a controller **628**. The controller **628** may be provided over the substrate **621** or at least partially embedded within the substrate **621**. The sensors **626** may be communicatively coupled to the controller **628** by electrically conductive interconnects **629**, such as copper traces, vias, pads, etc. In some embodiments, vias may pass through a thickness of the layer **622** and traces extend along a surface of the substrate **621** to provide the lateral connection to the controller **628**.

[0069] In an embodiment, the controller **628** may provide the processing, memory, and/or power for implementing the in-situ metrology with the optimization substrate **620**. For example, the controller **628** may comprise a processor, a memory, and a power source (e.g., a battery). Any suitable processor (e.g., central processing unit (CPU), application specific integrated circuit (ASIC), field programmable gate array (FPGA), etc.) may be used for the processor. Any suitable memory type may be used for the memory. The power source may be a rechargeable battery, such as a lithium ion battery, and/or the like.

[0070] In some embodiments, the data from the sensors **626** is stored on the memory within the controller **628**, and the memory is accessed after the optimization substrate **620** is removed from the tool. In other embodiments, the controller **628** may further comprise a wireless communications system (e.g., Wi-Fi, Bluetooth, etc.), in order to communicate with external devices in real time, in near real time, or within any desired time period.

[0071] Referring now to FIG. **7**, a plan view illustration of a cluster tool **730** is shown, in accordance with an embodiment. The cluster tool **730** may be a semiconductor processing tool that comprises chambers **734** for processes such as a such as, but not limited to, PVD, ALD, CVD, PEALD PECVD, etching, FEP, ECP, ion implantation, or the like. In an embodiment, the cluster tool **730** may comprise an EFEM **731**. The EFEM **731** may receive front opening unified pods (FOUPs) **736** or other substrate transport devices. A wafer handling robot within the EFEM **731** transfers substrates from the FOUP **736** to a load lock **732**. The load lock **732** is coupled to a transfer chamber **733** that is held at a vacuum pressure. That is, the load lock **732** allows for the transition from an atmospheric pressure environment to a vacuum environment. A substrate handling robot within the transfer chamber **733** can distribute wafers from the load lock **732** to any of the chambers **734** that are coupled to the transfer chamber **733**. For example, eight chambers **734** are shown in FIG. **7**. Though, one or more chambers **734** may be used in some embodiments. In an embodiment, a metrology tool **735** may be coupled to the cluster tool **730** as well. In the illustrated embodiment, the metrology tool **735** is coupled to the EFEM **731**. Other embodiments may include a metrology tool **735** that is coupled to the transfer chamber **733**. The metrology tool **735** may be used to provide the inline metrology described in process **200** in some embodiments.

[0072] In an embodiment, an optimization substrate (e.g., similar to any of the optimization substrates described in greater detail herein) may be provided in a FOUP **736** and transferred into the cluster tool **730** in a manner similar to any other substrate processed in the cluster tool **730**. The optimization substrate may be used to clean and/or perform in-situ metrology in any of the components of the cluster tool **730**. That is, the optimization substrate may clean and/or monitor one or more of the EFEM **731**, the load lock **732**, the transfer chamber **733**, the metrology tool **735**, and/or any of the chambers **734** in accordance with process **200** described in greater detail herein.

[0073] Referring now to FIG. **8**, a cross-sectional illustration of a tool **850** that may be monitored and cleaned with an optimization substrate **820** is shown, in accordance with an embodiment. In an embodiment, the tool **850** may comprise a chamber **851**. The chamber **851** may be suitable for supporting a vacuum pressure. For example, the vacuum pressure may be suitable to support the formation of a plasma **857** within the chamber **851**. Exhaust and pumping structures are omitted for simplicity. In an embodiment, a pedestal **854** may be provided in the chamber **851** for supporting a substrate (e.g., optimization substrate **820**). The substrate may be inserted into the chamber **851** through a slit valve **852** when a door **853** is opened. The substrate may be placed on lift pins **856** that can then lower (as indicated by the arrows) the substrate onto the pedestal **854**. The pedestal **854** may comprise a chucking architecture (e.g., an electrostatic chuck (ESC)). The pedestal **854** may be temperature controlled as well. For example, resistive heating elements **855** may be provided in the pedestal.

[0074] In an embodiment, a lid **858** may seal the chamber **851**. The lid **858** may include fluidic pathways **859** for flowing gas into **862** the lid **858** and out **861** into the chamber **851**. The lid **858** may also be coupled to a power source **863** (e.g., a radio frequency (RF) power source, a microwave power source, etc.), in order to couple power into the chamber **851** to ignite and/or sustain the plasma **857**.

[0075] In an embodiment, the optimization substrate **820** may comprise a substrate **821** and a layer **822**. The optimization substrate **820** may also comprise sensors (not shown). In an embodiment, the optimization substrate **820** may be similar to any of the optimization substrates described in greater detail herein. In an embodiment, the optimization substrate **820** may be used in order to clean the chamber **851** and to monitor one or more different process parameter within the tool **850** through use of in-situ metrology. For example, process parameters may include one or more of gas flow rates from the lid **858**, temperature of the pedestal **854**, properties of the plasma **857**, performance of the lift pins **856**, vibrations, particle concentrations, and/or the like. While FIG. **8** provides one exemplary tool **850** configuration, it is to be appreciated that process **200** and optimization substrates **820** may be used in accordance with any tool architecture and/or type.

[0076] Referring now to FIG. **9**, a block diagram of an exemplary computer system **900** of a processing tool is illustrated in accordance with an embodiment. In an embodiment, computer system **900** is coupled to and controls processing in the processing tool. Computer system **900** may be connected (e.g., networked) to other machines in a Local Area Network (LAN), an intranet, an extranet, or the Internet. Computer system **900** may operate in the capacity of a server or a client machine in a client-server network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. Computer system **900** may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated for computer system **900**, the term "machine" shall also be taken to include any collection of machines (e.g., computers) that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies described herein.

[0077] Computer system **900** may include a computer program product, or software **922**, having a non-transitory machine-readable medium having stored thereon instructions, which may be used to program computer system **900** (or other electronic devices) to perform a process according to embodiments. A machine-readable medium includes any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium (e.g., read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory devices, etc.), a machine (e.g., computer) readable transmission medium (electrical, optical, acoustical or other form of propagated signals (e.g., infrared signals, digital signals, etc.)), etc.

[0078] In an embodiment, computer system **900** includes a system processor **902**, a main memory **904** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **906** (e.g., flash memory, static random access memory (SRAM), etc.), and a secondary memory **918** (e.g., a data storage device), which communicate with each other via a bus **930**.

[0079] System processor **902** represents one or more general-purpose processing devices such as a microsystem processor, central processing unit, or the like. More particularly, the system processor may be a complex instruction set computing (CISC) microsystem processor, reduced instruction set computing (RISC) microsystem processor, very long instruction word (VLIW) microsystem processor, a system processor implementing other instruction sets, or system processors implementing a combination of instruction sets. System processor **902** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal system processor (DSP), network system processor, or the like. System processor **902** is configured to execute the processing logic **926** for performing the operations described herein.

[0080] The computer system **900** may further include a system network interface device **908** for communicating with other devices or machines. The computer system **900** may also include a video display unit **910** (e.g., a liquid crystal display (LCD), a light emitting diode display (LED), or a cathode ray tube (CRT)), an alphanumeric input device **912** (e.g., a keyboard), a cursor control device **914** (e.g., a mouse), and a signal generation device **916** (e.g., a speaker).

[0081] The secondary memory **918** may include a machine-accessible storage medium **931** (or more specifically a computer-readable storage medium) on which is stored one or more sets of instructions (e.g., software **922**) embodying any one or more of the methodologies or functions described herein. The software **922** may also reside, completely or at least partially, within the main memory **904** and/or within the system processor **902** during execution thereof by the computer system **900**, the main memory **904** and the system processor **902** also constituting machine-readable storage media. The software **922** may further be transmitted or received over a network **961** via the system network interface device **908**. In an embodiment, the network interface device **908** may operate using RF coupling, optical coupling, acoustic coupling, or inductive coupling.

[0082] While the machine-accessible storage medium **931** is shown in an exemplary embodiment to be a single medium, the term "machine-readable storage medium" should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term "machine-readable storage medium" shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies. The term "machine-readable storage medium" shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

[0083] In the foregoing specification, specific exemplary embodiments have been described. It will be evident that various modifications may be made thereto without departing from the scope of the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A method for maintaining a processing tool, comprising:

performing a planned maintenance (PM) process on the processing tool;

processing substrates on the processing tool;

performing an in-situ metrology and cleaning process on the processing tool with an optimization substrate to determine if a processing parameter of the processing tool is within a specified range; and

restarting substrate processing on the processing tool when the processing parameter is within the specified range.

2. The method of claim **1**, wherein the processing parameter is a particle concentration, and wherein the in-situ metrology and cleaning process comprises:

providing the optimization substrate to the processing tool, wherein the optimization substrate comprises:

a substrate;

a layer over the substrate, wherein the layer is configured to attract particles in the processing tool; and

a particle sensor on the substrate; and

processing the optimization substrate with an optimization recipe.

3. The method of claim **2**, wherein the particle sensor is a vibration sensor, a resonant structure sensor, or a laser light scattering sensor.

4. The method of claim **2**, wherein the optimization substrate is heated.

5. The method of claim **2**, wherein the optimization substrate is charged.

6. The method of claim **2**, wherein the optimization recipe is different than a process recipe used to process substrates on the processing tool.

7. The method of claim **1**, wherein the in-situ metrology and cleaning process comprises cycling the optimization substrate through the processing tool a plurality of times, and wherein the cycling is continued until the processing parameter is within the specified range.

8. The method of claim **1**, wherein the in-situ metrology and cleaning process is started after a predetermined duration, wherein the predetermined duration is a number of substrates that have been processed that is up to 200 substrates, or wherein the predetermined duration is a time duration that is up to one week.

9. The method of claim **1**, further comprising a validation process before processing substrates on the processing tool after the PM, wherein the validation process, comprises:

performing the in-situ metrology and cleaning process with the optimization substrate; and

performing a metrology process to confirm the processing tool is processing parameter is within the specified range before processing substrates.

10. The method of claim **9**, wherein the metrology process is inline metrology.

11. An apparatus, comprising:

a substrate with a first surface and a second surface opposite from the first surface;

a layer on the second surface of the substrate, wherein the layer is configured to attract particles from an environment surrounding the apparatus;

a plurality of sensors on the substrate, wherein the plurality of sensors are configured to detect particles landing on the apparatus; and

a controller on the substrate and communicatively coupled to the plurality of sensors, wherein the controller comprises:

a processor;

a memory; and

a power source.

12. The apparatus of claim **11**, wherein the layer comprises a pattern, and wherein the pattern comprises one or more trenches into a surface of the layer.

13. The apparatus of claim **11**, wherein the layer comprises one or more of silicon nitride, silicon oxide, aluminum, titanium, a polymer, or a sol-gel based material.

14. The apparatus of claim **11**, further comprising:

a second layer on the first surface of the substrate.

15. The apparatus of claim **11**, wherein the substrate comprises one or more of silicon, a ceramic, silicon carbide, graphite, or aluminum.

16. The apparatus of claim **11**, wherein the substrate has a wafer form factor or a panel form factor.

17. The apparatus of claim **11**, wherein the plurality of sensors comprise one or more of an accelerometer, a resonant structure, or a laser light scattering sensor.

18. A method for maintaining a processing tool, comprising:

performing a planned maintenance (PM) process on the processing tool;

processing substrates on the processing tool;

performing an in-situ metrology and cleaning process on the processing tool with an optimization substrate to determine if a processing parameter of the processing tool is within a specified range, wherein the in-situ metrology and cleaning process comprises:

providing the optimization substrate to the processing tool, wherein the optimization substrate comprises:

a substrate;

a layer over the substrate, wherein the layer is configured to attract particles in the processing tool; and

a sensor on the substrate configured to detect the processing parameter;

processing the optimization substrate with an optimization recipe a plurality of cycles until the processing parameter is within the specified range; and

restarting substrate processing on the processing tool when the processing parameter is within the specified range.

19. The method of claim **18**, wherein the processing tool is a cluster tool.

20. The method of claim **18**, wherein the processing parameter is one or more of a particle concentration, a vibration amount, a temperature, a gas flow, or an optical reading.

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